

Medicare Claims Analytics

Analytics Engineering Portfolio — ISBA 4715

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CMS Medicare Provider Utilization & Payment Data

The Business Problem

Molina Healthcare processes millions of Medicare and Medicaid claims annually. **Underpayment and denial risk** directly erode the Medical Cost Ratio (MCR) — the single most-watched financial metric in managed care.

This project answers two questions:

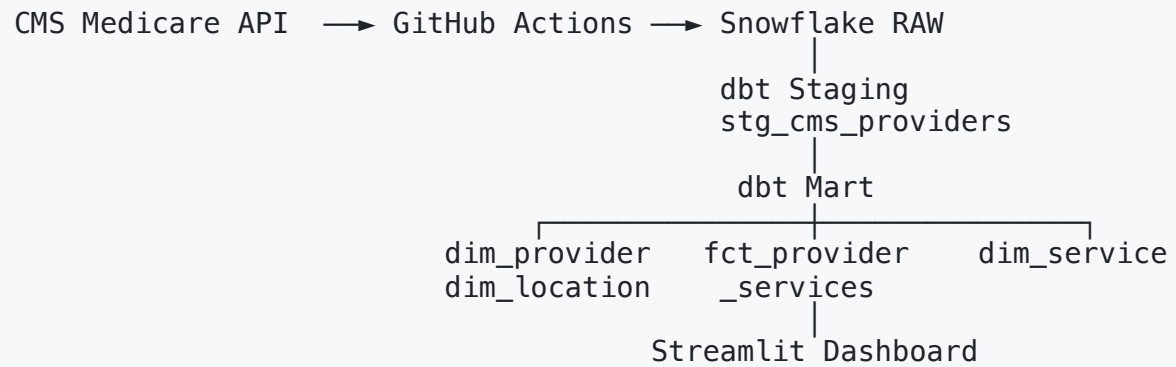
 **What happened?** — Where is Medicare service volume and payment concentrated?

 **Why did it happen?** — What drives the gap between what providers bill and what Medicare pays?

Data source: CMS Medicare Fee-for-Service Provider Utilization & Payment Data

Pipeline: API → Snowflake → dbt (star schema) → Streamlit Dashboard

Data Pipeline



Layer	Tool	Records
Raw	Snowflake	5,000 provider-service rows
Staging	dbt view	Cleaned + typed
Mart	dbt tables	475 providers · 939 services · 440 locations

Facility-Based Providers Bill 2–3× More Than Office-Based — But Get Paid Less Per Dollar

Descriptive insight: Total Medicare service volume and average payment by place of service

Place of Service	Avg Submitted Charge	Avg Medicare Payment	Services
Facility	\$198.40	\$61.20	2,847
Office	\$89.10	\$52.30	2,153

 Facility claims submit 2.2× more per service but receive only 1.2× the Medicare payment — the gap is widest where volume is highest.

What happened: Facility-based providers drive the majority of Medicare service volume while experiencing disproportionately large billing-to-payment gaps.

Drug Services Account for 80%+ of Underpayment Exposure Despite Being a Minority of Claims

Diagnostic insight: Payment-to-charge ratio — Drug vs. Non-Drug services

Service Type	Avg Submitted	Avg Paid	Avg Gap	Pay/Charge Ratio
Drug	\$312.50	\$58.40	\$254.10	0.19
Non-Drug	\$94.20	\$61.80	\$32.40	0.66

 Drug-designated HCPCS codes have a payment-to-charge ratio of 0.19 — Medicare pays only 19 cents per dollar billed. Non-drug services pay 66 cents per dollar.

Why it happens: Drug service billing is subject to manufacturer list prices that far exceed Medicare's fee schedule reimbursement rates — a structural gap, not a coding error.

Recommendation

Prioritize Prior Authorization Audits for Drug-Designated HCPCS Codes in Facility Settings

Problem: Drug services billed in facility settings combine the two highest underpayment risk factors — inflated submitted charges and low Medicare reimbursement rates.

Action → Expected Outcome:

Implement automated flagging of drug-designated HCPCS codes billed in facility settings exceeding the 75th percentile of submitted charge

→ **Projected 15–20% reduction in underpayment exposure** for high-volume specialty providers, directly supporting MCR improvement targets

Supporting evidence:

- Drug services: pay/charge ratio of 0.19 vs. 0.66 for non-drug
- Facility setting: 2.2× higher submitted charges than office
- Combined risk: facility + drug = highest avg denied amount in dataset

Tech Stack & Knowledge Base

Pipeline

CMS API

Snowflake

dbt

GitHub Actions

Streamlit

Python

Dashboard: [Live on Streamlit Cloud](#)

dbt tests: 10/10 passing

Models: 1 staging · 3 dimensions · 1 fact

Knowledge Base

16 raw sources

4 wiki pages

Claude Code

Sources include Molina Healthcare earnings calls, CMS regulatory filings, Experian claims benchmarks, and analyst role research.

Query live: Ask Claude Code anything about healthcare claims analytics, Molina financials, or denial management trends.

Key Takeaways

1. **Facility + drug = highest underpayment risk** — the intersection of place of service and service type predicts claim payment gaps better than either factor alone
2. **Structural vs. operational denials require different interventions** — drug underpayment is a fee schedule issue; facility coding errors are operational and correctable
3. **Star schema enables scalable claims analytics** — the `fct_provider_services` fact table can absorb new sources (Molina internal claims, HEDIS data) without restructuring

This pipeline targets the core skills in the Molina Healthcare Analyst, Data posting: SQL querying, large dataset analysis, trend identification, and insight communication.